

ADVANCED QUANTUM MECHANICS

Many body physics



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LECTURE 12: Creation/annihilation and field operators

Let's review main aspects of the last lecture. Writing at the level of wavefunctions we have the distinction between two types of particles which behave very differently

Bosons versus fermions

To fulfill the symmetry conditions, two-particle states built from single-particle orbitals ϕ_a, ϕ_b must be symmetrized (bosons) or antisymmetrized (fermions).

Bosons. The symmetrized spatial wavefunction is

$$\psi_B(\mathbf{r}_1, \mathbf{r}_2) = \begin{cases} \frac{1}{\sqrt{2}} [\phi_a(\mathbf{r}_1)\phi_b(\mathbf{r}_2) + \phi_a(\mathbf{r}_2)\phi_b(\mathbf{r}_1)], & a \neq b, \\ \phi_a(\mathbf{r}_1)\phi_a(\mathbf{r}_2), & a = b, \end{cases}$$

so bosons *can* occupy the same single-particle state.

Fermions. The antisymmetric fermionic state is the Slater determinant

$$\psi_F(\mathbf{r}_1, \mathbf{r}_2) = \frac{1}{\sqrt{2}} \begin{vmatrix} \phi_a(\mathbf{r}_1) & \phi_b(\mathbf{r}_1) \\ \phi_a(\mathbf{r}_2) & \phi_b(\mathbf{r}_2) \end{vmatrix} = \frac{1}{\sqrt{2}} [\phi_a(\mathbf{r}_1)\phi_b(\mathbf{r}_2) - \phi_b(\mathbf{r}_1)\phi_a(\mathbf{r}_2)],$$

which vanishes identically if $\phi_a = \phi_b$, expressing the Pauli exclusion principle.

In the following we will proceed by first focusing on bosons and going through:

- (i) creation and annihilation operators
- (ii) field operators
- (iii) Hamiltonian expansion both in terms of field operators and in terms of creation/annihilation operators
- (iv) discuss two-body interactions
- (v) give some examples

We will then go through the same steps for fermions.

For bosons

First we construct single-particle states from the eigenstates of the single-particle Hamiltonian. For each eigenstate of $\hat{h}$ with energy E_j and wavefunction ϕ_j , we denote the abstract ket $|1_j\rangle$. The Hamiltonian is then diagonal,

$$\hat{h} = \sum_j E_j |1_j\rangle\langle 1_j|.$$

In the symmetrized two-particle basis, the bosonic symmetry condition written earlier at the level of wavefunctions can be expressed now at the level of kets:

$$|1_j, 1_{j'}\rangle_B = \begin{cases} \frac{1}{\sqrt{2}}(|1_j\rangle |1_{j'}\rangle + |1_{j'}\rangle |1_j\rangle), & j \neq j', \\ |1_j\rangle |1_j\rangle, & j = j'. \end{cases}$$

Acting with the two-particle Hamiltonian gives

$$\hat{H} |1_j, 1_{j'}\rangle_B = (E_j + E_{j'}) |1_j, 1_{j'}\rangle_B.$$

Now we introduce the occupation number (or Fock state) description. We start by defining the *vacuum state*

$$|0\rangle \equiv |0_0, 0_1, 0_2, \dots, 0_j, \dots\rangle,$$

which means that no bosons occupy any of the single-particle eigenstates $|1_j\rangle$ of $\hat{h}$.

To create or remove particles in these eigenstates, we introduce *bosonic creation and annihilation operators* $\hat{a}_j^\dagger$ and $\hat{a}_j$ with the commutation relations

$$[\hat{a}_j, \hat{a}_{j'}^\dagger] = \delta_{jj'}, \quad [\hat{a}_j, \hat{a}_{j'}] = [\hat{a}_j^\dagger, \hat{a}_{j'}^\dagger] = 0.$$

The vacuum is annihilated by all $\hat{a}_j$:

$$\hat{a}_j |0\rangle = 0.$$

Acting once with $\hat{a}_j^\dagger$ creates one boson in the single-particle eigenstate j :

$$|1\rangle_j \equiv |0_0, 0_1, \dots, 1_j, \dots\rangle = \hat{a}_j^\dagger |0\rangle.$$

Repeated action produces n_j bosons in eigenstate j :

$$|n_j\rangle_j \equiv |0_0, 0_1, \dots, n_j, \dots\rangle = \frac{(\hat{a}_j^\dagger)^{n_j}}{\sqrt{n_j!}} |0\rangle.$$

To have N bosonic particles in the system means to place $N = n_0 + n_1 + \dots + n_j$ onto the states of the single particle Hamiltonian. Below is an example for a system with two particles.

Two-boson Fock states

For a system with *two bosons*, one may place both bosons in the same mode:

$$|2_0, 0_1, 0_2, \dots\rangle = \frac{(\hat{a}_0^\dagger)^2}{\sqrt{2!}} |0\rangle, \quad |0_0, 2_1, 0_2, \dots\rangle = \frac{(\hat{a}_1^\dagger)^2}{\sqrt{2!}} |0\rangle \quad \dots$$

Or we may place the two bosons in *different* modes:

$$|1_0, 1_1, 0_2, \dots\rangle = \hat{a}_0^\dagger \hat{a}_1^\dagger |0\rangle, \quad |1_0, 0_1, 1_2, 0_3, \dots\rangle = \hat{a}_0^\dagger \hat{a}_2^\dagger |0\rangle \quad \dots$$

Let's check how one can translate back from the Fock (occupation number) notation to recover wavefunctions. For example for 2 particles in the same state,

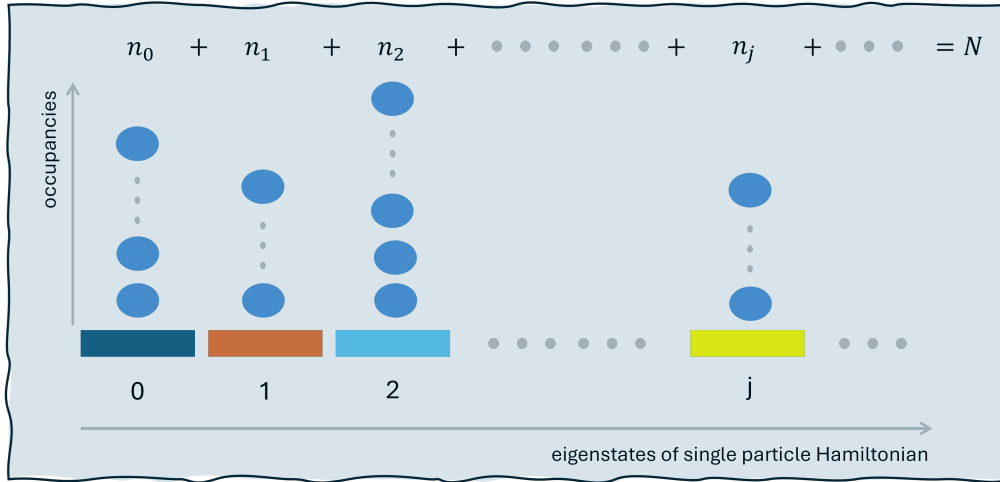
$$\langle x_1, x_2 | \underbrace{|2_0, 0_1, 0_2, \dots\rangle}_{\text{Fock space}} = \langle x_1, x_2 | 1_0, 1_0 \rangle_B = \phi_0(x_1)\phi_0(x_2).$$

For another state, where particles are in different states one computes

$$\langle x_1, x_2 | \underbrace{|1_0, 0_1, 1_2, \dots\rangle}_{\text{Fock space}} = \langle x_1, x_2 | 1_0, 1_2 \rangle_B = \frac{1}{\sqrt{2}} [\phi_0(x_1)\phi_2(x_2) + \phi_2(x_1)\phi_0(x_2)].$$

which is manifestly symmetric under $x_1 \leftrightarrow x_2$. The important observation is that one has to use the fact that the states in the ket notation are defined already in a symmetrized way.

BOSONS



Let us now notice that for any arbitrary bosonic Fock state

$$|n_0, n_1, \dots\rangle = \prod_j \frac{(\hat{a}_j^\dagger)^{n_j}}{\sqrt{n_j!}} |0\rangle.$$

the following action is just a number operator action

$$\hat{a}_j^\dagger \hat{a}_j |n_0, n_1, \dots\rangle = n_j |n_0, n_1, \dots\rangle,$$

pulling out the number of particles in that particular state. Since a state with occupation numbers $\{n_j\}$ (over all possible j 's) has energy $E = \sum_j n_j E_j$, the operator reproducing this on all states is

$$\hat{H} = \sum_j E_j \hat{a}_j^\dagger \hat{a}_j.$$

This is the way of writing the Hamiltonian fully with creation and annihilation operators. The situation here is applicable for Bose-Einstein condensation of bosons onto the ground state of an externally imposed potential or for superconductivity.

Introducing the field operator - 1D treatment

The creation operators create particles in the eigenstates of the Hamiltonian. This is similar to when we look at a plane wave, which is a fully delocalized state with a fixed momentum. In second quantization we may also describe the creation or annihilation of a boson directly at a point x . For this we introduce the *bosonic field operators*

$$\hat{\Psi}(x), \quad \hat{\Psi}^\dagger(x),$$

which destroy or create a particle at position x . These operators are expanded in the eigenbasis $\{\phi_j(x)\}$ of the single-particle Hamiltonian as

$$\hat{\Psi}(x) = \sum_j \phi_j(x) \hat{a}_j, \quad \hat{\Psi}^\dagger(x) = \sum_j \phi_j^*(x) \hat{a}_j^\dagger.$$

Their action is then

$$\hat{\Psi}^\dagger(x) |0\rangle = \sum_j \phi_j^*(x) |1\rangle_j,$$

leading to a superposition of all eigenstates of the single particle Hamiltonian with the proper weights.

Because the operators $\hat{a}_j, \hat{a}_j^\dagger$ satisfy the canonical commutation relations, the field operators obey

$$[\hat{\Psi}(x), \hat{\Psi}^\dagger(x')] = \delta(x - x'), \quad [\hat{\Psi}(x), \hat{\Psi}(x')] = 0, \quad [\hat{\Psi}^\dagger(x), \hat{\Psi}^\dagger(x')] = 0.$$

For the nontrivial commutation above, we can prove it

$$\begin{aligned} [\hat{\Psi}(x), \hat{\Psi}^\dagger(x')] &= \left[\sum_j \phi_j(x) \hat{a}_j, \sum_k \phi_k^*(x') \hat{a}_k^\dagger \right] = \sum_{j,k} \phi_j(x) \phi_k^*(x') [\hat{a}_j, \hat{a}_k^\dagger] \\ &= \sum_{j,k} \phi_j(x) \phi_k^*(x') \delta_{jk} = \sum_j \phi_j(x) \phi_j^*(x') = \delta(x - x'). \end{aligned}$$

Above, we used the property that the single particle wavefunctions are a complete set on the x axis. Now we use the fact that they are orthonormal,

$$\int dx \phi_{j'}^*(x) \phi_j(x) = \delta_{j'j}.$$

to invert the relation. Multiplying $\hat{\Psi}(x)$ by $\phi_{j'}^*(x)$ and integrating over x gives

$$\hat{a}_j = \int dx \phi_j^*(x) \hat{\Psi}(x), \quad \hat{a}_j^\dagger = \int dx \phi_j(x) \hat{\Psi}^\dagger(x).$$

Starting from

$$\hat{H} = \sum_j E_j \hat{a}_j^\dagger \hat{a}_j,$$

we substitute the expressions for $\hat{a}_j^\dagger$ and $\hat{a}_j$:

$$\hat{a}_j^\dagger \hat{a}_j = \left(\int dx \phi_j(x) \hat{\Psi}^\dagger(x) \right) \left(\int dx' \phi_j^*(x') \hat{\Psi}(x') \right).$$

Thus,

$$\hat{H} = \sum_j E_j \int dx \int dx' \phi_j(x) \phi_j^*(x') \hat{\Psi}^\dagger(x) \hat{\Psi}(x').$$

Reordering the sums and operators gives

$$\hat{H} = \int dx \int dx' \hat{\Psi}^\dagger(x) \underbrace{\left[\sum_j E_j \phi_j(x) \phi_j^*(x') \right]}_{\hat{h}(x) \delta(x - x')} \hat{\Psi}(x').$$

Finally we can write

$$\hat{H} = \int dx \hat{\Psi}^\dagger(x) \left(-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} + V(x) \right) \hat{\Psi}(x),$$

or simply

$$\hat{H} = \int dx \hat{\Psi}^\dagger(x) \hat{h}(x) \hat{\Psi}(x).$$

For fermions

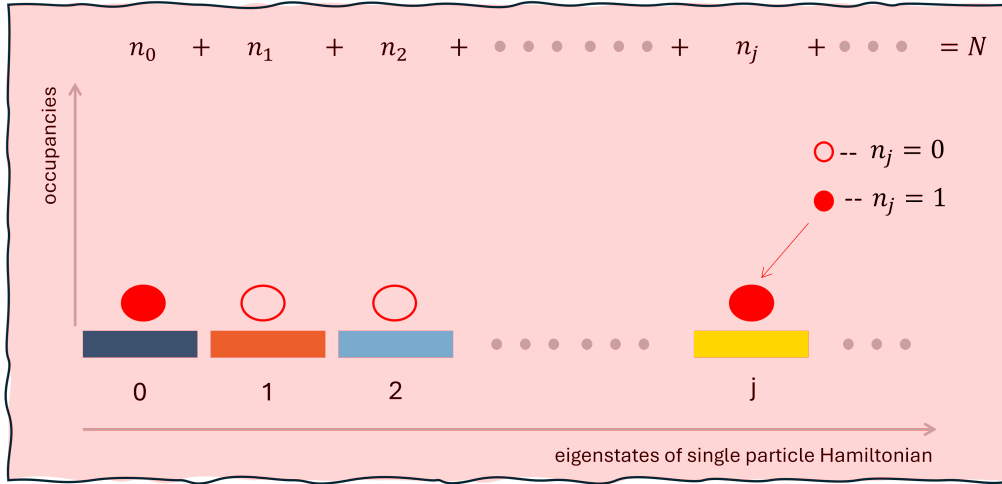
We now describe the same system for *fermions*, again using occupation numbers instead of explicit two-particle wavefunctions. As before, we introduce the *vacuum state*

$$|0\rangle \equiv |0_0, 0_1, 0_2, \dots, 0_j, \dots\rangle.$$

To create or remove particles in these eigenstates, we now introduce *fermionic creation and annihilation operators* $\hat{c}_j^\dagger$ and $\hat{c}_j$ satisfying the canonical anticommutation relations

$$\{\hat{c}_j, \hat{c}_{j'}^\dagger\} = \delta_{jj'}, \quad \{\hat{c}_j, \hat{c}_{j'}\} = \{\hat{c}_j^\dagger, \hat{c}_{j'}^\dagger\} = 0.$$

FERMIONS



The vacuum is annihilated by all $\hat{c}_j$:

$$\hat{c}_j |\mathbf{0}\rangle = 0.$$

Acting once with $\hat{c}_j^\dagger$ creates one fermion in the eigenstate j :

$$|1\rangle_j \equiv |0_0, 0_1, \dots, 1_j, \dots\rangle = \hat{c}_j^\dagger |\mathbf{0}\rangle.$$

Due to the Pauli exclusion principle, each single-particle level can host *at most one fermion*. Therefore, $(\hat{c}_j^\dagger)^2 = 0$.

To have N fermionic particles in the system means choosing N distinct occupied modes. Below is the corresponding example for a system with two fermions.

Two-fermion Fock states

For a system with *two fermions*, the Pauli principle requires that they occupy *different* single-particle states. A few possible configurations are

$$|1, 1, 0, 0, \dots\rangle = \hat{c}_0^\dagger \hat{c}_1^\dagger |\mathbf{0}\rangle, \quad |1, 0, 1, 0, \dots\rangle = \hat{c}_0^\dagger \hat{c}_2^\dagger |\mathbf{0}\rangle, \quad |0, 1, 1, 0, \dots\rangle = \hat{c}_1^\dagger \hat{c}_2^\dagger |\mathbf{0}\rangle.$$

States such as $|2, 0, 0, \dots\rangle$ or $|0, 2, 0, \dots\rangle$ do *not* exist because two fermions cannot occupy the same single-particle state.

For any fermionic Fock state

$$|n_0, n_1, \dots\rangle = \prod_j (\hat{c}_j^\dagger)^{n_j} |\mathbf{0}\rangle, \quad n_j \in \{0, 1\},$$

the number operator acts as

$$\hat{c}_j^\dagger \hat{c}_j |n_0, n_1, \dots\rangle = n_j |n_0, n_1, \dots\rangle.$$

Since a state with occupations $\{n_j\}$ has energy $E = \sum_j n_j E_j$, the Hamiltonian becomes

$$\hat{H} = \sum_j E_j \hat{c}_j^\dagger \hat{c}_j.$$

Fermionic field operators

Define the fermionic field operators

$$\hat{\Psi}(x) = \sum_j \phi_j(x) \hat{c}_j, \quad \hat{\Psi}^\dagger(x) = \sum_j \phi_j^*(x) \hat{c}_j^\dagger,$$

with anticommutation relations

$$\{\hat{\Psi}(x), \hat{\Psi}^\dagger(x')\} = \delta(x - x'), \quad \{\hat{\Psi}(x), \hat{\Psi}(x')\} = 0.$$

Orthonormality allows inversion:

$$\hat{c}_j = \int dx \phi_j^*(x) \hat{\Psi}(x), \quad \hat{c}_j^\dagger = \int dx \phi_j(x) \hat{\Psi}^\dagger(x).$$

Substituting into the mode Hamiltonian yields

$$\hat{H} = \sum_j E_j \hat{c}_j^\dagger \hat{c}_j = \int dx \hat{\Psi}^\dagger(x) \hat{h}(x) \hat{\Psi}(x).$$

For the explicit single-particle operator

$$\hat{h}(x) = -\frac{\hbar^2}{2m} \frac{d^2}{dx^2} + V(x),$$

the Hamiltonian becomes

$$\hat{H} = \int dx \hat{\Psi}^\dagger(x) \left(-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} + V(x) \right) \hat{\Psi}(x).$$

Intermission - we need to deal with two-body interactions

Let us now explicitly refer to an interesting system, which is the Helium atom. Here one solves the problem of two electrons orbiting a nucleus.

The Helium atom

The single-particle Hamiltonian is described by the Hamiltonian made up of kinetic energy of the electron and the Coulomb attraction

$$\hat{h} = -\frac{\hbar^2}{2m} \nabla^2 - \frac{Ze^2}{4\pi\epsilon_0 |\mathbf{r}|},$$

with eigenfunctions $\phi_j(\mathbf{r})$ and eigenvalues E_j (known as this is a hydrogen-like problem with $Z = 2$ instead of 1).

Placing the two electrons together means that we have to antisymmetrize the problem.

However, we are still missing something as the electrons also interact via the Coulomb repulsion so that the full Hamiltonian is

$$\hat{H} = \hat{h}_1 + \hat{h}_2 + \hat{V}(\mathbf{r}_1 - \mathbf{r}_2),$$

with the Coulomb repulsion included as potential term symmetric under exchange of particles

$$V(\mathbf{r}_1 - \mathbf{r}_2) = \frac{e^2}{4\pi\epsilon_0 |\mathbf{r}_1 - \mathbf{r}_2|}.$$

To solve the problem, we use the two particle eigenbasis antisymmetrized as for fermions $|1_j, 1_k\rangle_F$. Thus the Hamiltonian becomes

$$\hat{H} = \sum_{j,k} (E_j + E_k) |1_j, 1_k\rangle_F \langle 1_j, 1_k| + \sum_{j,k,\ell,m} V_{jk\ell m} |1_j, 1_k\rangle_F \langle 1_\ell, 1_m|.$$

and the matrix elements of the interaction are

$$V_{jk\ell m} = \iint d^3r_1 d^3r_2 \phi_j^*(\mathbf{r}_1) \phi_k^*(\mathbf{r}_2) V(\mathbf{r}_1 - \mathbf{r}_2) \phi_\ell(\mathbf{r}_1) \phi_m(\mathbf{r}_2).$$

The intermission above is only meant to justify that one needs to pay attention to the interaction as well and learn how to deal with it both in terms of creation/annihilation operators as well as with field operators.

Two-body interactions in second quantization - the case for bosons

Up to now the Hamiltonian contained only the single-particle contribution

$$\hat{H}_1 = \sum_j E_j \hat{a}_j^\dagger \hat{a}_j = \sum_{j,k} (E_j + E_k) |1_j, 1_k\rangle_B \langle 1_j, 1_k|,$$

where we used the two-particle eigenbasis symmetrized for bosons,

$$|1_j, 1_k\rangle_B = \frac{1}{\sqrt{2}} (|1_j\rangle |1_k\rangle + |1_k\rangle |1_j\rangle), \quad (j \neq k).$$

To include interactions, we write the potential operator $\hat{V}(x_1 - x_2)$ in matrix form in this basis, obtaining the full two-particle Hamiltonian

$$\hat{H} = \sum_{j,k} (E_j + E_k) |1_j, 1_k\rangle_B \langle 1_j, 1_k| + \sum_{j,k,\ell,m} V_{jk\ell m} |1_j, 1_k\rangle_B \langle 1_\ell, 1_m|.$$

The matrix elements of the interaction are now 1D integrals,

$$V_{jk\ell m} = \iint dx_1 dx_2 \phi_j^*(x_1) \phi_k^*(x_2) V(x_1 - x_2) \phi_\ell(x_1) \phi_m(x_2).$$

Rewriting in second quantization. To express the same physics directly in terms of occupation numbers, insert two resolutions of the identity,

$$\mathbb{1} = \int dx |x\rangle \langle x|,$$

and expand the position ket in the eigenbasis of $\hat{h}$,

$$|x\rangle = \sum_j \phi_j(x) |1_j\rangle, \quad \phi_j(x) = \langle x | 1_j \rangle.$$

Acting with the interaction operator on a symmetrized two-boson state gives

$$\hat{V} |1_\ell, 1_m\rangle_B = \sum_{j,k} V_{jk\ell m} |1_j, 1_k\rangle_B.$$

Thus, in the Fock (occupation number) representation the interaction Hamiltonian becomes

$$\hat{H}_{\text{int}} = \frac{1}{2} \sum_{j,k,\ell,m} V_{jk\ell m} \hat{a}_j^\dagger \hat{a}_k^\dagger \hat{a}_m \hat{a}_\ell,$$

where the factor 1/2 avoids double counting of boson pairs.

Putting the single-particle and interaction parts together, we obtain the standard second-quantized form

$$\hat{H} = \sum_j E_j \hat{a}_j^\dagger \hat{a}_j + \frac{1}{2} \sum_{j,k,\ell,m} V_{jk\ell m} \hat{a}_j^\dagger \hat{a}_k^\dagger \hat{a}_m \hat{a}_\ell.$$

Field-operator form in 1D. Define the bosonic field operators via

$$\hat{\Psi}(x) = \sum_j \phi_j(x) \hat{a}_j, \quad \hat{\Psi}^\dagger(x) = \sum_j \phi_j^*(x) \hat{a}_j^\dagger,$$

with the completeness relation

$$\sum_j \phi_j(x) \phi_j^*(x') = \delta(x - x').$$

The single-particle Hamiltonian becomes

$$\hat{H}_1 = \int dx \hat{\Psi}^\dagger(x) \left(-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} + V(x) \right) \hat{\Psi}(x).$$

For the interaction, insert the field expansions into

$$\hat{H}_{\text{int}} = \frac{1}{2} \int dx_1 \int dx_2 \hat{\Psi}^\dagger(x_1) \hat{\Psi}^\dagger(x_2) V(x_1 - x_2) \hat{\Psi}(x_2) \hat{\Psi}(x_1),$$

and use completeness to recover the matrix-element form above.

Final 1D Hamiltonian for interacting bosons:

$$\hat{H} = \int dx \hat{\Psi}^\dagger(x) \left(-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} + V(x) \right) \hat{\Psi}(x) + \frac{1}{2} \int dx_1 dx_2 \hat{\Psi}^\dagger(x_1) \hat{\Psi}^\dagger(x_2) V(x_1 - x_2) \hat{\Psi}(x_2) \hat{\Psi}(x_1).$$

This is the 1D starting point for the Gross–Pitaevskii description once a contact potential $V(x_1 - x_2) = g_{1D} \delta(x_1 - x_2)$ is assumed.

Two-body interactions for fermions

For spinless fermions in one dimension, the two-particle eigenbasis is

$$|1_j, 1_k\rangle_F = \frac{1}{\sqrt{2}} (|1_j\rangle|1_k\rangle - |1_k\rangle|1_j\rangle), \quad (j \neq k),$$

with the understanding that

$$|1_j, 1_j\rangle_F = 0,$$

reflecting the Pauli exclusion principle.

Including a two-body interaction potential $V(x_1 - x_2)$, the full two-particle Hamiltonian reads

$$\hat{H} = \sum_{j,k} (E_j + E_k) |1_j, 1_k\rangle_F \langle 1_j, 1_k| + \sum_{j,k,\ell,m} V_{jk\ell m} |1_j, 1_k\rangle_F \langle 1_\ell, 1_m|.$$

The 1D matrix elements are

$$V_{jk\ell m} = \iint dx_1 dx_2 \phi_j^*(x_1) \phi_k^*(x_2) V(x_1 - x_2) \phi_\ell(x_1) \phi_m(x_2),$$

and automatically include the antisymmetry of the fermionic basis.

Using the fermionic creation and annihilation operators

$$\{\hat{a}_j, \hat{a}_k^\dagger\} = \delta_{jk}, \quad \{\hat{a}_j, \hat{a}_k\} = 0,$$

the full Hamiltonian for interacting fermions in 1D is

$$\hat{H} = \sum_j E_j \hat{a}_j^\dagger \hat{a}_j + \frac{1}{2} \sum_{j,k,\ell,m} V_{jk\ell m} \hat{a}_j^\dagger \hat{a}_k^\dagger \hat{a}_m \hat{a}_\ell$$

Introduce fermionic field operators

$$\hat{\Psi}(x) = \sum_j \phi_j(x) \hat{a}_j, \quad \hat{\Psi}^\dagger(x) = \sum_j \phi_j^*(x) \hat{a}_j^\dagger,$$

which satisfy

$$\{\hat{\Psi}(x), \hat{\Psi}^\dagger(x')\} = \delta(x - x'), \quad \{\hat{\Psi}(x), \hat{\Psi}(x')\} = 0.$$

Then the Hamiltonian becomes

$$\hat{H} = \int dx \hat{\Psi}^\dagger(x) \left(-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} + V(x) \right) \hat{\Psi}(x) + \frac{1}{2} \int dx_1 dx_2 \hat{\Psi}^\dagger(x_1) \hat{\Psi}^\dagger(x_2) V(x_1 - x_2) \hat{\Psi}(x_2) \hat{\Psi}(x_1).$$

This is the standard 1D second-quantized Hamiltonian for interacting spinless **fermions**. Important differences from the bosonic case:

- $\hat{c}_a^\dagger \hat{c}_b^\dagger = -\hat{c}_b^\dagger \hat{c}_a^\dagger$, so V_{abcd} effectively acts only on antisymmetric combinations.
- $(\hat{c}_j^\dagger)^2 = 0$, so no two fermions can occupy the same single-particle state.
- The factor 1/2 again removes double-counting of identical pairs.

Contact interaction - towards Gross-Pitaevski for BEC in 1D

We describe a weakly interacting Bose gas in one dimension by bosonic field operators $\hat{\Psi}(x), \hat{\Psi}^\dagger(x)$. The single-particle Hamiltonian induces the second-quantized many-body Hamiltonian

$$\hat{H}_0 = \int dx \hat{\Psi}^\dagger(x) \hat{h}(x) \hat{\Psi}(x) = \int dx \hat{\Psi}^\dagger(x) \left(-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} + V(x) \right) \hat{\Psi}(x).$$

At low temperatures and low densities, a dilute Bose gas is dominated by *binary collisions* at very low relative momenta. In this regime, the microscopic interaction potential $V_{\text{mic}}(r)$ between two atoms can be replaced by a much simpler *pseudopotential* that reproduces only the low-energy scattering properties.

In three dimensions, for short-range interactions and low collision energies, only the *s*-wave ($\ell = 0$) scattering channel contributes. The detailed shape of $V_{\text{mic}}(r)$ is then irrelevant, and one introduces the contact (delta) pseudopotential

$$V_{3D}(\mathbf{r} - \mathbf{r}') = g_{3D} \delta^{(3)}(\mathbf{r} - \mathbf{r}'),$$

with coupling constant

$$g_{3D} = \frac{4\pi\hbar^2 a_s}{m},$$

where a_s is the *s*-wave scattering length and m is the particle mass. This pseudopotential generates the same *s*-wave scattering amplitude as the true potential at low energies.

In a strongly elongated (quasi-1D) geometry with tight harmonic confinement in the transverse directions, the many-body dynamics along the longitudinal axis can be described by an effective 1D model. Integrating out the frozen transverse degrees of freedom leads to an effective 1D contact interaction

$$U(x - x') = g_{1D} \delta(x - x'),$$

where g_{1D} is the effective 1D interaction strength (which depends on a_s and the transverse confinement). In our 1D description we simply take this as the starting point.

From first quantization to field operators.

In first quantization, for N identical bosons in 1D, the interaction Hamiltonian with the contact potential reads

$$\hat{H}_{\text{int}} = \frac{1}{2} \sum_{i \neq j} U(x_i - x_j) = \frac{g_{1D}}{2} \sum_{i \neq j} \delta(x_i - x_j),$$

where the factor $1/2$ avoids double counting of pairs (i, j) and (j, i) .

In second quantization, a general two-body interaction potential $U(x - x')$ is written in terms of field operators as

$$\hat{H}_{\text{int}} = \frac{1}{2} \int dx \int dx' \hat{\Psi}^\dagger(x) \hat{\Psi}^\dagger(x') U(x - x') \hat{\Psi}(x') \hat{\Psi}(x).$$

To see this, expand the field operators in any single-particle basis $\{\phi_j(x)\}$,

$$\hat{\Psi}(x) = \sum_j \phi_j(x) \hat{a}_j, \quad \hat{\Psi}^\dagger(x) = \sum_j \phi_j^*(x) \hat{a}_j^\dagger,$$

insert these into the integral expression, and use the orthonormality of the $\phi_j(x)$ together with the bosonic commutation relations to recover the first-quantized form

$$\hat{H}_{\text{int}} = \frac{1}{2} \sum_{i \neq j} U(x_i - x_j).$$

For the 1D contact potential $U(x - x') = g_{1D} \delta(x - x')$ we obtain

$$\hat{H}_{\text{int}} = \frac{1}{2} \int dx \int dx' \hat{\Psi}^\dagger(x) \hat{\Psi}^\dagger(x') g_{1D} \delta(x - x') \hat{\Psi}(x') \hat{\Psi}(x).$$

Performing the x' -integral using the delta function gives

$$\hat{H}_{\text{int}} = \frac{g_{1D}}{2} \int dx \hat{\Psi}^\dagger(x) \hat{\Psi}^\dagger(x) \hat{\Psi}(x) \hat{\Psi}(x),$$

which is the interaction Hamiltonian used in the Gross–Pitaevskii description for a 1D Bose gas in the dilute, low-energy, s -wave scattering regime. The full Hamiltonian is

$$\hat{H} = \hat{H}_0 + \hat{H}_{\text{int}}.$$

The Heisenberg equation of motion for the field operator reads

$$i\hbar \frac{\partial}{\partial t} \hat{\Psi}(x, t) = [\hat{\Psi}(x, t), \hat{H}].$$

Using the commutation relations and integrating by parts, one obtains

$$[\hat{\Psi}(x), \hat{H}_0] = \left(-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} + V(x) \right) \hat{\Psi}(x),$$

and a short commutator algebra for the interaction term gives

$$[\hat{\Psi}(x), \hat{H}_{\text{int}}] = g_{1D} \hat{\Psi}^\dagger(x) \hat{\Psi}(x) \hat{\Psi}(x).$$

Thus the exact Heisenberg equation becomes

$$i\hbar \frac{\partial}{\partial t} \hat{\Psi}(x, t) = \left(-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} + V(x) \right) \hat{\Psi}(x, t) + g_{1D} \hat{\Psi}^\dagger(x, t) \hat{\Psi}(x, t) \hat{\Psi}(x, t).$$

This calculation is left for the exercise session.

For a Bose–Einstein condensate we introduce the mean-field (order parameter)

$$\psi(x, t) \equiv \langle \hat{\Psi}(x, t) \rangle,$$

assume that all particles occupy (approximately) the same single-particle state (Hartree ansatz), and factorize operator products as

$$\langle \hat{\Psi}^\dagger(x, t) \hat{\Psi}(x, t) \hat{\Psi}(x, t) \rangle \approx |\psi(x, t)|^2 \psi(x, t).$$

Taking the expectation value of the Heisenberg equation and replacing $\hat{\Psi}(x, t) \rightarrow \psi(x, t)$ gives the time-dependent Gross–Pitaevskii equation in 1D:

$$i\hbar \frac{\partial}{\partial t} \psi(x, t) = \left(-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} + V(x) + g_{1D} |\psi(x, t)|^2 \right) \psi(x, t).$$

The condensate wavefunction is normalized to the total particle number,

$$\int dx |\psi(x, t)|^2 = N,$$

so that $|\psi(x, t)|^2$ is the 1D particle density.